Mode S and the Kivenlahti Mast: the Lowest 300 m at Helsinki-Vantaa

Aircraft winds and temperatures against a 327-m weather mast, the MEPS model and the radiosonde

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Abstract. MODE-S Wind derives winds and temperatures from the Mode S Enhanced Surveillance replies of the aircraft around Helsinki-Vantaa (EFHK) and describes the airport's boundary layer hour by hour — the wind in 250-ft bins and the temperature in 200-ft bands from the landings and departures. Since 1 November 2026 it also collects the observations of the Finnish Meteorological Institute's 327-m weather mast at Kivenlahti, Espoo, 24 km south-west of the airport: temperature at eight and wind at four levels between 2 and 327 m, every 10 minutes. This log compares the two, with the MEPS model and the Jokioinen radiosonde as third and fourth sources, over the first 7.4 days (25 October 20 UTC – 2 November 06 UTC 2026, 179 hours). At 217–297 m the aircraft temperatures agree with the mast within 0.1 K (RMS 0.56–0.65 K); lower down they are colder (−0.47 K at 92 m). The aircraft's temperature difference between 297 m and the surface follows the mast's with a correlation of 0.83, so the hourly stability of the lowest 300 m is seen; shallow night inversions below about 100 m are mostly not. The wind speeds correlate at 0.92–0.95; the Mode S, MEPS and sonde winds all differ from the mast's in the same way at two of its levels, which points to the mast's own exposure rather than to the aircraft.

Research only. A passive, uncertified receiver; a mast 24 km from the runways; one autumn week. Nothing here is an official observation, and none of it may be used for flight operations or air traffic control.

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About this document

This is a *living* study log, written for aviation meteorologists and atmospheric scientists, in the same form as the windshear study log [5]. Each later check of the stored data gets its own dated entry (Section 6); earlier results are not rewritten, a correction is marked as such. Times are UTC. Heights are metres above the local ground unless said otherwise (the mast's ground is 44 m above mean sea level, the airport's 179 ft = 55 m); "ft MSL" is used where the pages use it. Differences are always *source* − *mast*.

Table 1. Version history of this document.

Version	Date	Content
1.0	2 Nov 2026	First edition: the mast and its integration, the first 7.4 days (179 hours) against Mode S, MEPS and the soundings — temperature, stability and inversions, wind; a case of a shallow evening inversion (29 October); limits and next steps.

1 Setting and data

1.1 The Kivenlahti mast

The mast (FMI station *Espoo Latokaski*, fmisid 101000, 60.178 N 24.640 E) stands on ground 44 m above sea level in the Kivenlahti district of Espoo, a few kilometres from the coast of the Gulf of Finland and 23.6 km south-west of EFHK, roughly on the line of runways 04L/04R extended. It is 327 m tall and measures every 10 minutes the **temperature, relative humidity and dewpoint at 2, 26, 49, 92, 141, 217, 265 and 297 m** and the **wind speed, direction and gust at 26, 92, 217 and 327 m** above its ground. The data are FMI open data (CC BY 4.0 [1]), read from the WFS stored query `fmi::observations::weather::mast::multipointcoverage`.

In MODE-S Wind the collector `collector/mast.py` reads the new steps every 10 minutes (about 5 minutes after their time), back-fills up to 7 days after a start or a break, and keeps every step (table `mast_obs`, about 300 bytes a step, 15 MB a year). It was installed on 1 November 2026; the first start back-filled 1 037 steps, so the stored series begins on 25 October 2026 20:20 UTC. On the snapshot of 2 November 06:07 UTC it held 1 067 steps **without a gap**.

1.2 The other sources

- **Mode S boundary layer** (`bl_hours`, `collector/boundary.py` [5]): every hour the median wind (u, v) of all aircraft within 25 km of the airport in 250-ft bins from 250 ft MSL ($\geq$ 3 observations), and the temperature of the landings and departures in 200-ft bands from 400 ft MSL ($\geq$ 4 flights, smoothed over $\pm$200 ft), with the METAR temperature. The aircraft temperature is the Mach/TAS temperature of BDS 5,0 and 6,0 [2].
- **MEPS** [3]: the model levels of the newest run at the hour (the area mean around the receiver), heights above the ground from the hypsometric equation with the model's surface pressure; and the 2-m temperature and 10-m wind of the weather maps at the mast.
- **Jokioinen soundings** (about 100 km north-west, inland): the 00, 06 and 12 UTC launches, heights above the station's ground.

1.3 Matching

All comparisons are made **at the same height above the local ground** (Figure 1). Mode S: the hour's bins and bands at the airport (ft MSL $-$ 179 ft) interpolated to each mast level — winds through u and v across gaps up to 260 ft, temperatures across gaps up to 420 ft — against the mast's *hour mean* (temperature the mean of the six steps, wind the vector mean). The lowest Mode S temperature band (400 ft MSL) is 67 m above the airport, so the mast's 2, 26 and 49 m have no aircraft temperature; its 26-m wind has an aircraft value in only 15 hours. MEPS: the model at the full hour against the mast step at that time. Soundings: the step nearest the launch. The tool is `web/api/mast.py` (`compare`, `modes`), the same code as the Sounding page's Mast tab and its time series.

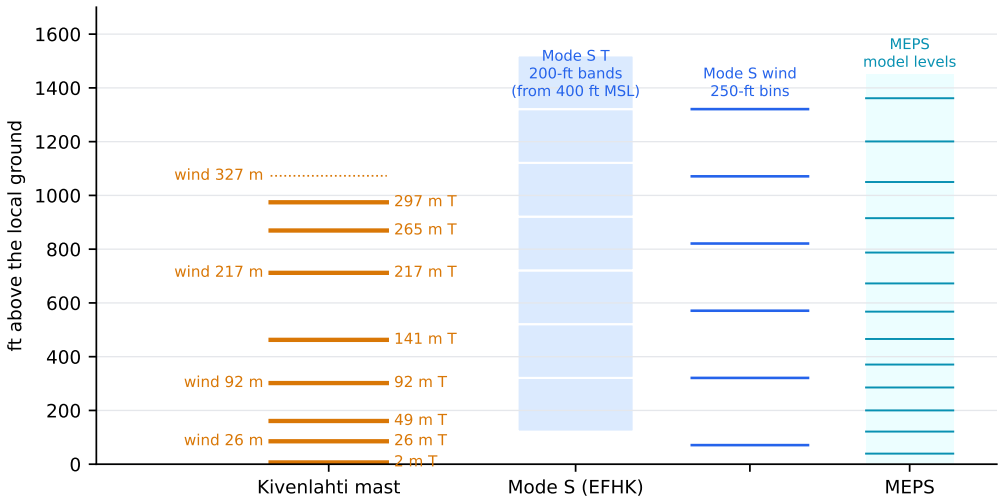


Figure 1. Heights of the compared values above the local ground: the mast's eight temperature and four wind levels, the Mode S temperature bands and wind bins at the airport (from 400 and 250 ft MSL), and the MEPS model levels of 1 November 2026 05 UTC.

1.4 The period

25 October 20 UTC – 2 November 06 UTC 2026, 179 hours (snapshot of 2 November 06:07 UTC). Mode S has a boundary-layer value at the mast's heights in 147 of them (temperature at 297 m); the gap is the night: between 00 and 03 UTC only 1 of 24 hours had enough flights. MEPS in 177 hours, soundings 22. The weather (Figure 2): a southerly flow of 25–40 kt at 327 m and 9–12 °C at first, weakening and cooling after 28 October; an evening with a calm surface under a fresh flow aloft on 29 October (Section 3.3); and calm high pressure on 31 October – 1 November (327-m wind 1–17 kt, the METAR wind 0–3 kt overnight).

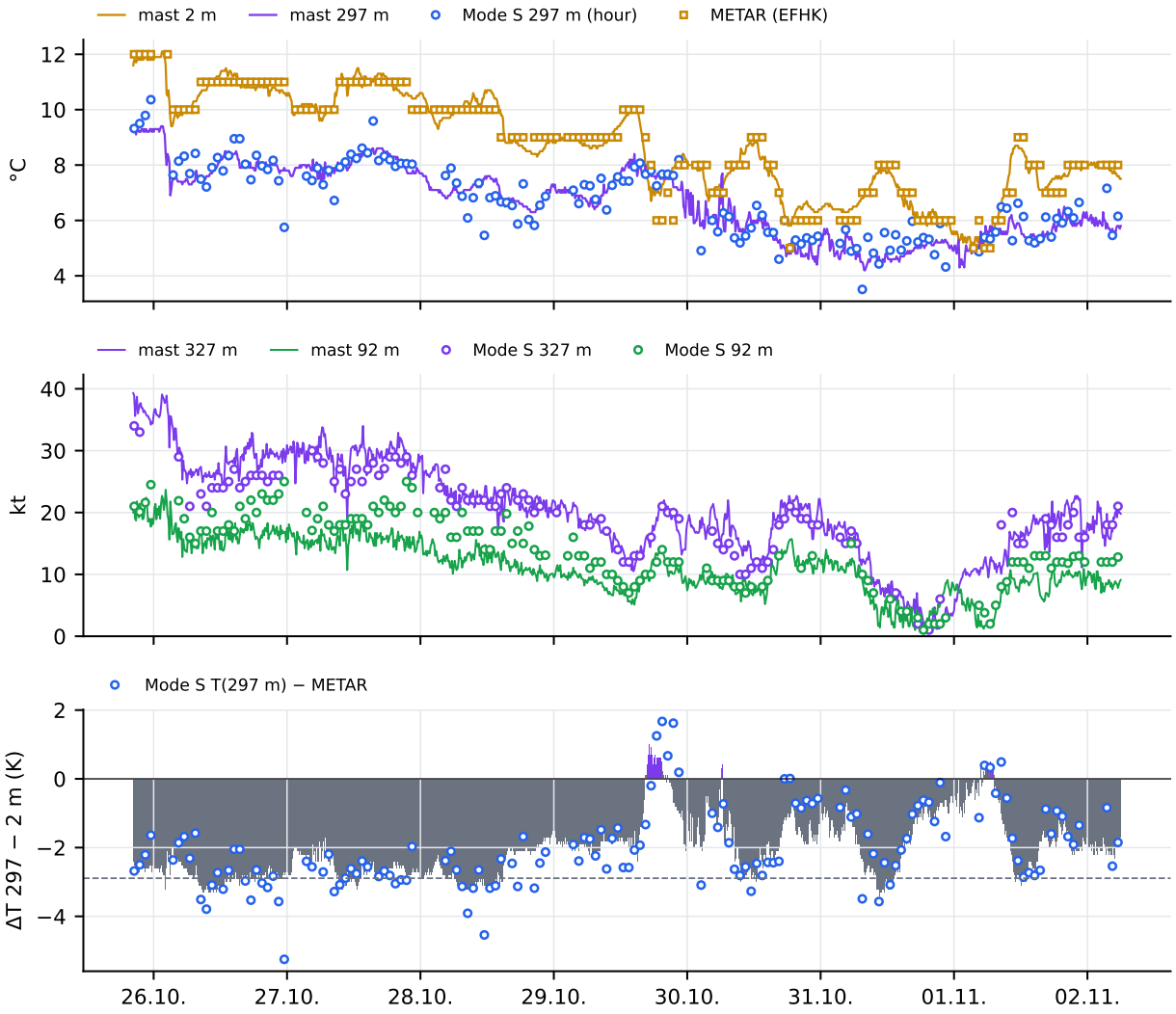


Figure 2. The period. Top: the mast's 2-m and 297-m temperature (10-minute steps), the Mode S temperature at 297 m above the airport (hour) and the EFHK METAR. Middle: wind speed at 327 and 92 m, mast and Mode S. Bottom: the mast's temperature difference 297 − 2 m (violet above +0.3 K; dashed: dry-adiabatic −2.9 K) and the Mode S equivalent, T(297 m) − METAR.

2 Temperature

Table 2. Temperature, source − mast: mean difference / RMS (K) and the number of hours (soundings: launches), 25 Oct 20 UTC – 2 Nov 06 UTC 2026.

Mast level	Mode S (EFHK)	MEPS model levels	Jokioinen sonde
297 m	+0.06 / 0.56 (147)	+0.05 / 0.36 (177)	−0.29 / 0.82 (22)
265 m	−0.08 / 0.61 (147)	−0.04 / 0.38 (177)	−0.46 / 0.90 (22)
217 m	−0.10 / 0.65 (147)	0.00 / 0.42 (177)	−0.47 / 0.89 (22)
141 m	−0.33 / 0.82 (146)	−0.12 / 0.48 (177)	−0.74 / 1.02 (22)
92 m	−0.47 / 0.97 (135)	−0.11 / 0.50 (177)	−0.71 / 0.95 (22)
49 m	—	−0.05 / 0.51 (177)	−0.66 / 0.90 (22)
26 m	—	+0.01 / 0.54 (177)	−0.58 / 0.83 (22)

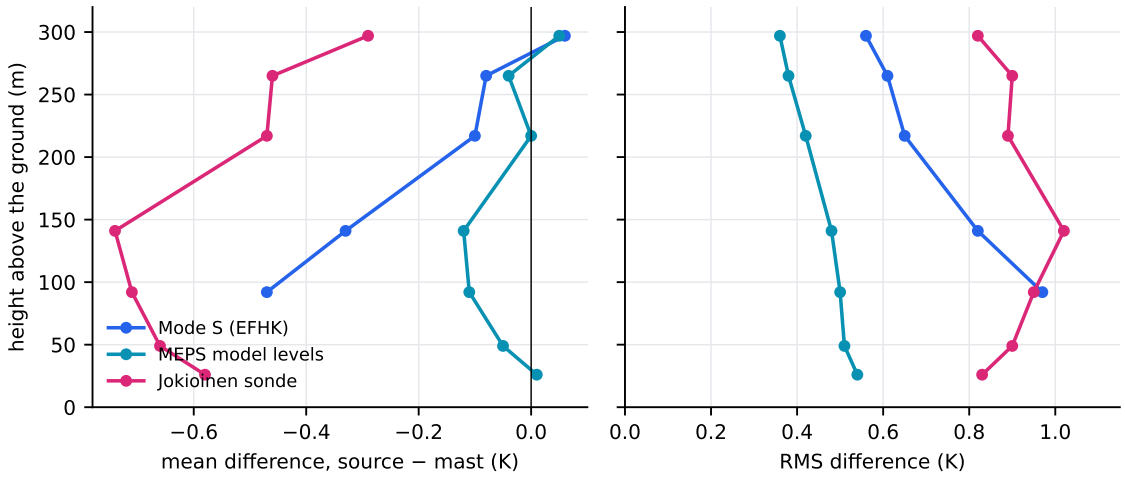


Figure 3. Temperature difference to the mast by level: mean (left) and RMS (right), Mode S, MEPS and the Jokioinen sonde.

- **Mode S at 217–297 m agrees with the mast within 0.1 K**, RMS 0.56–0.65 K — about the size of the hour-to-hour scatter that two places 24 km apart may have. The correlation at 297 m is 0.91 (Figure 8).
- **Below 200 m the aircraft are colder**: −0.33 K at 141 m, −0.47 K at 92 m, and the RMS grows to 0.97 K. The difference is the same by day (06–15 UTC: −0.51 K at 92 m) and by night (−0.44 K), so it is not the land–sea contrast of the night alone.
- **MEPS** agrees with the mast at every level within 0.12 K (RMS 0.36–0.54 K) — the model of the area around the receiver, not of the mast's site, is as close to the mast as Mode S is at the top and closer below.
- **The sonde** is 0.3–0.7 K colder: Jokioinen is 100 km inland, and most of the 22 launches were at 00 and 06 UTC.

The cause of the Mode S cold difference near the ground is not yet known. Candidates: (1) the lowest bands rest on the last part of the approaches and the first of the climb-outs, and the ±200-ft smoothing reaches down to the 400-ft band only from above; (2) the airport and its surroundings may really be colder than the coastal mast in the lowest 100 m (the METAR was 1–2 K colder than the mast's 2 m in calm evenings, Section 3.3) — but MEPS, an area mean around the airport, does not show it; (3) an effect of the flight phase on the Mach/TAS temperature, as the phase offset of the airspeed found in the windshear study [5]. Separating landings from departures, and runway by runway, is the next check (Section 7).

3 Stability and inversions

3.1 The temperature difference of the lowest 300 m

Mode S has no temperature at 2 m, but the METAR temperature of the airport is its surface value. The difference T(297 m) − METAR follows the mast's T(297 m) − T(2 m) with a **correlation of 0.83** over 143 hours, a mean difference of +0.07 K and an RMS of 0.66 K (Figure 4, left). The daily cycle is the same in both: the median close to dry-adiabatic (−2.6 to −2.9 K) at 09–14 UTC and least negative in the evening, 18–20 UTC (Figure 4, right). MEPS has the same cycle between its 297 and 26 m (correlation 0.72 with the mast's, RMS 0.48 K), but flatter in the evening.

Within the layer the aircraft do less well: their difference between 297 and 92 m correlates with the mast's at only 0.35, and its mean is −1.26 K against the mast's −1.73 K. The aircraft see the layer's mean temperature and its difference to the ground, not the lapse inside the lowest 200 m — consistent with the cold difference at 92 m.

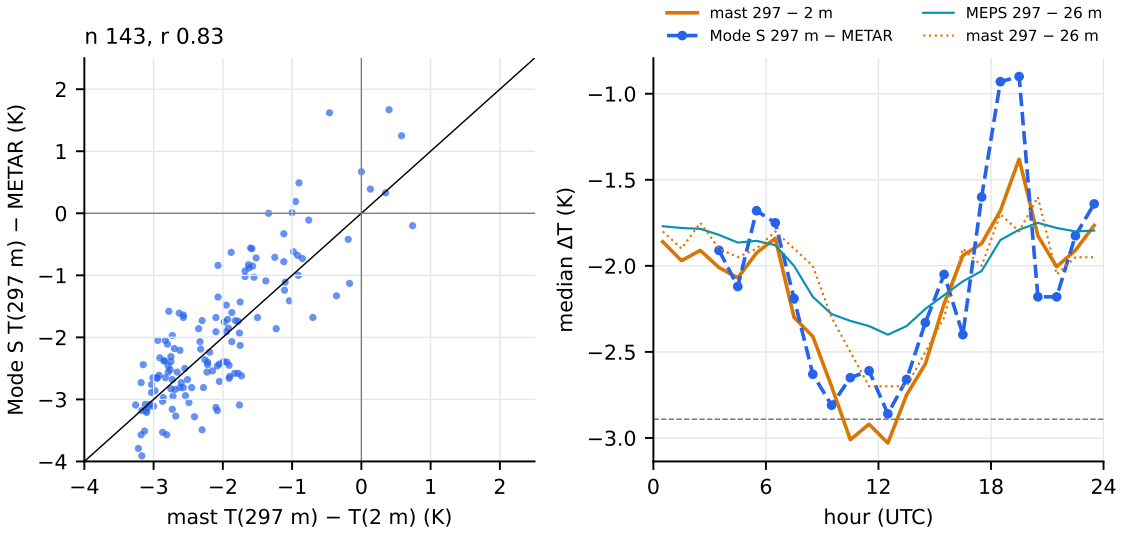


Figure 4. Left: the mast's T(297 m) − T(2 m) against Mode S T(297 m) − METAR, hour by hour. Right: median by hour of the day, with the MEPS and mast 297 − 26 m (dashed grey: dry-adiabatic over 295 m).

3.2 Inversions

With T(297) − T(2) above +0.3 K the mast had an inversion over the whole 300 m in only 4 hours. The inversions of the period were *shallow*: measured as the warmest level minus the 2 m, the mast had ≥ 0.5 K in **24 of 179 hours** (13 %), ≥ 1 K in 10 and ≥ 2 K in 3, with the warmest level mostly at 49–141 m (Figure 5). All were in two spells: the evening of 29 October (up to 2.5 K) and the calm night of 31 October – 1 November (0.5–1.2 K).

Table 3. The mast's inversions (≥ 0.5 K, warmest level − 2 m) against the Mode S surface inversion of the boundary-layer hours (the 600–1 000-ft air at least 0.5 K warmer than the METAR). Hours with a Mode S value.

	Mode S inversion	no Mode S inversion
Mast inversion	7	14
No mast inversion	1	142

The Mode S criterion found 7 of the 21 mast inversion hours that had a Mode S hour — the stronger ones of 29 October 18–22 UTC and of 1 November 05–06 UTC — and raised one hour without a mast inversion. The ones missed were weak (0.5–1.2 K) and had their warmest level at 26–92 m: below the lowest Mode S band and below the criterion's 600–1 000 ft. Three more hours of the second night (1 November 00–02 UTC) had no Mode S hour at all. The mast thus adds the shallow surface inversions that the aircraft cannot see; a lower Mode S criterion (the 400–600-ft bands against the METAR) is a question for the next check.

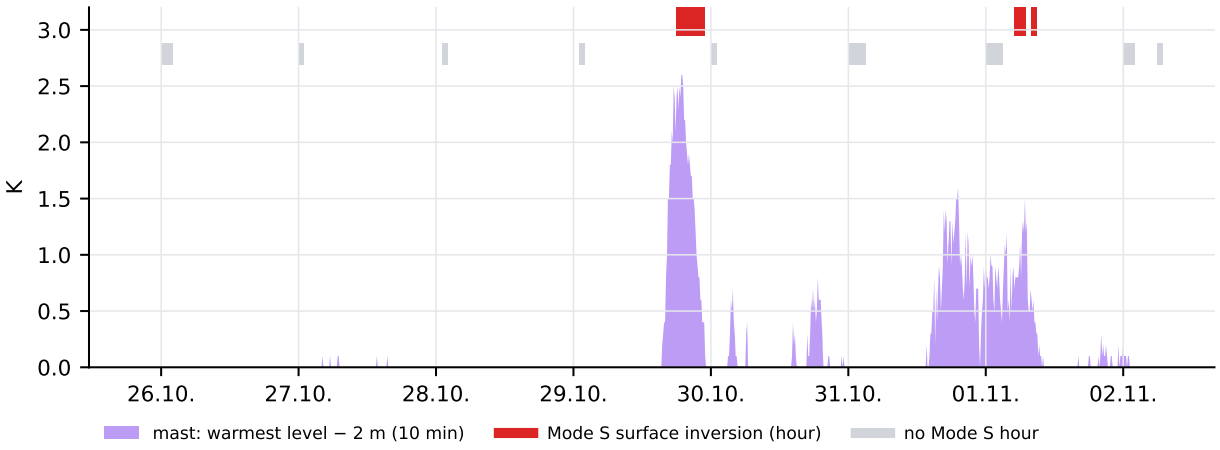


Figure 5. The mast's inversion strength (warmest level − 2 m, 10-minute steps) with the hours of a Mode S surface inversion (red) and the hours without a Mode S value (grey).

3.3 A shallow evening inversion: 29 October 2026

In the afternoon and evening the wind at the ground fell to 4–5 kt while 327 m kept 15–21 kt. At the mast the 2-m temperature fell from 9.9 to 6.8 °C between 14 and 18 UTC while 92 m stayed at 9.2–9.5 °C: a 2.5-K inversion in the lowest 100 m, with the 297-m air (7.6 °C) again colder than 92 m above it (Figure 6). At the airport the METAR fell to 6 °C; the aircraft at 92 m above the airport read 7.1–8.0 °C and the boundary-layer hours flagged a surface inversion of +0.9 to +1.9 K from 18 to 22 UTC with its top at 1 000–1 600 ft MSL. **MEPS** kept its 26-m level at 8.8–9.0 °C through the evening — 0.7–0.9 K warmer than the mast at 18–21 UTC. Over the whole period the model's 26-m temperature was 0.35–0.5 K too warm in the hours with a mast inversion and right (−0.08 K) in the others: the model does not cool the lowest layer enough on calm evenings, a known difficulty of stable boundary layers in numerical models.

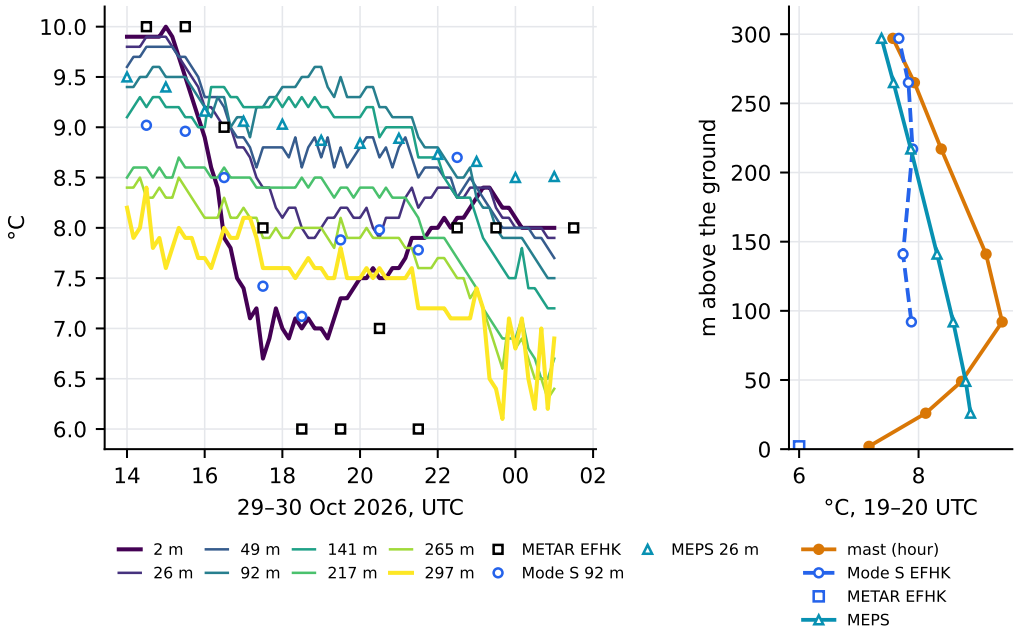


Figure 6. 29 October 2026, 14 – 01 UTC. Left: the mast's temperatures at its eight levels (10-minute steps), the EFHK METAR, Mode S at 92 m above the airport (hour) and MEPS at 26 m. Right: the profiles of 19–20 UTC — mast, Mode S with the METAR, and MEPS.

4 Wind

Table 4. Wind, source − mast: speed difference (kt) / vector RMS (kt) / direction difference (°, both ≥ 5 kt), and the number of hours (soundings: launches).

Mast level	Mode S (EFHK)	MEPS model levels	Jokioinen sonde
327 m	−1.3 / 4.6 / −8 (115)	−1.6 / 6.2 / −13 (177)	−0.6 / 7.2 / −10 (22)
217 m	+0.9 / 5.8 / +16 (127)	+1.0 / 5.8 / +12 (177)	+0.7 / 7.1 / +10 (22)
92 m	+2.8 / 4.0 / 0 (138)	+2.6 / 4.6 / −1 (177)	0.0 / 4.4 / −6 (22)
26 m	(+7.7 / 8.5; 15 hours)	+4.5 / 5.5 / +4 (177)	+0.5 / 2.6 / −4 (22)

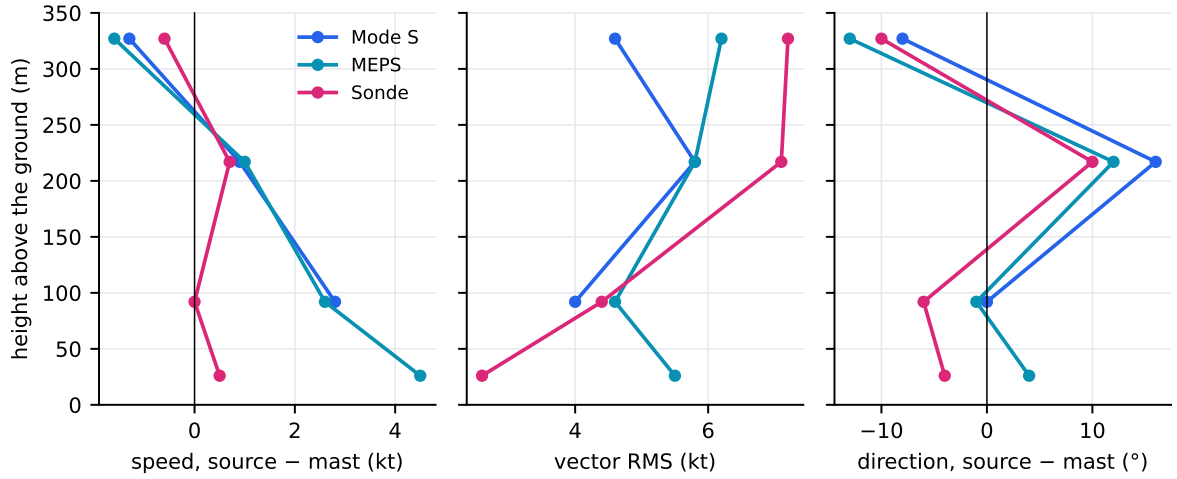


Figure 7. Wind difference to the mast by level: speed, vector RMS and direction (levels with at least 20 values).

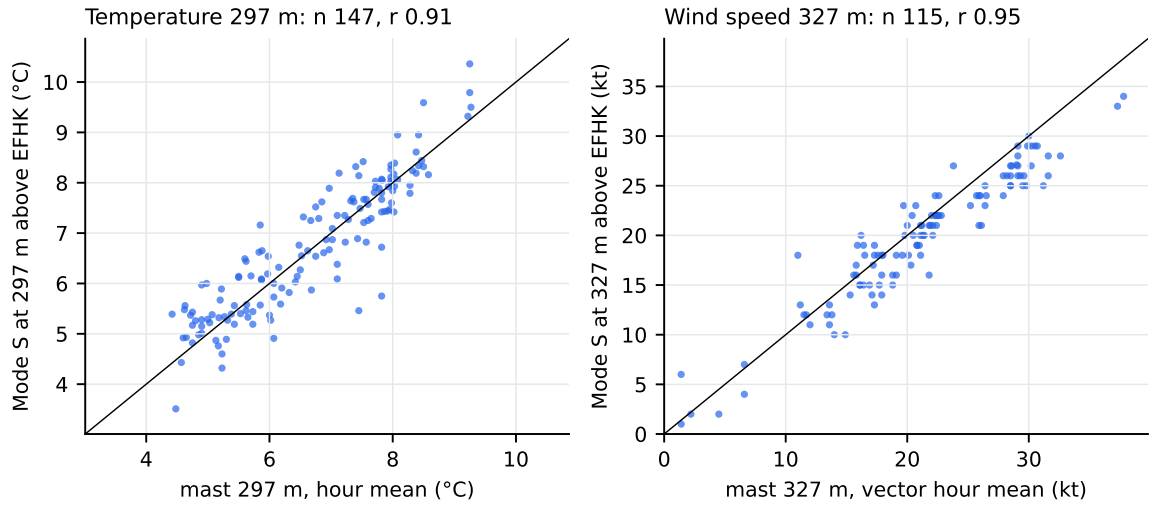


Figure 8. Mode S against the mast hour by hour: temperature at 297 m (left) and wind speed at 327 m (right).

- **The speeds follow each other closely**: correlation 0.92 (92 m), 0.95 (217 m) and 0.95 (327 m) for Mode S, 0.89–0.96 for MEPS.
- **The directions point at the mast.** At 217 m all three sources are 10–16° veered from the mast, at 327 m 8–13° backed; at 92 m they agree within 6°. Three independent sources — aircraft at the airport, a model and a balloon 100 km away — agreeing on opposite offsets at two neighbouring levels is unlikely to be the weather; the likeliest causes are the orientation of the mast's 217-m and 327-m wind sensors or the flow around the mast at those booms. To be checked with FMI's station metadata.
- **The speed profile**: Mode S and MEPS have 2.6–2.8 kt more wind than the mast at 92 m and 1.3–1.6 kt less at 327 m, so the mast's wind increases more with height (92 → 327 m: 9.8 kt on average, Mode S 5.7 kt). The sonde, at an open inland site, has none of the 92-m difference. Part of it may be the mast's surroundings slowing its lowest levels, part the airport's more open fetch; MEPS shows the same as Mode S because its surface is the area mean. At 26 m the few Mode S values (15 hours) and MEPS (+4.5 kt) read much more wind than the mast — the 26-m level is clearly in the mast site's own roughness layer.
- **Low-level jet**: Mode S found a jet in 9 hours of the period (28–29 October and 2 November, south-westerly, nose at 1 250–2 000 ft MSL, 19–25 kt; 30 October a northerly one at 750–1 000 ft). The nose was above the mast's top in the south-westerly cases, so the mast saw the increase below it (17–22 kt at 327 m, 4–6 kt at 26 m). On 30 October 18 UTC, with the northerly nose at about 170 m above the airport, the mast had nearly the same speed at 217 and 327 m (20.9 and 21.2 kt) — the increase with height stopped at about the jet's height.

5 What the mast adds

- **The layer below the aircraft**: temperatures at 2–49 m and the wind at 26 m, where the boundary-layer hours have nothing (the lowest band 67 m, the wind at 26 m in 15 of 179 hours). The shallow evening and night inversions of this period were all there.
- **Continuity**: a value every 10 minutes, also between 00 and 03 UTC when only one of 24 hours had enough flights.
- **A reference for the aircraft temperatures near the ground** and for MEPS in stable conditions (Sections 2 and 3.3).
- **A check of the inversion and jet criteria** of the boundary-layer hours, which so far had only the METAR below the aircraft.

5.1 Where it is shown

The Sounding page's **Mast** tab (tiles, profile 0–1 500 ft with Mode S, MEPS and the sonde, time series with the Mode S hours and the mast's 1-h mean, the comparison tables), the **Time–height** section (the mast's own strip and its columns inside the boundary-layer fields), the **Skew-T**, the Live Map's **EFHK now** box (the mast rows with their difference to MEPS and Mode S), the Windshear page's **BL** chart and the phone's **Mast** view — see the Sounding page guide [6] and the README.

6 Database checks

Date	Check and main results
2 Nov 2026	**First comparison** (snapshot 2 Nov 06:07 UTC, logs and soundings of the same morning). *Health*: the mast fetcher started with every restart of 1–2 November and back-filled the missing steps (1 037 at the first start, then 0–2); 1 067 steps 25 Oct 20:20 – 2 Nov 06:00 UTC without a gap; no error from the mast in the logs. The other warnings of the night were the receiver feed's usual night silences, one failed MEPS request and one failed pilot-report request (both retried). *Results*: Sections 2–4. ⇒ open questions in Section 7.

7 Limits, open questions and next steps

7.1 Limits

- **One autumn week** with few stable hours (24 with an inversion) and no front, convection or sea breeze; the numbers will change.
- **Distance**: the mast is 24 km from the runways and near the coast; in calm weather the two places really differ. The comparison measures the sum of the aircraft's error and that difference.
- **Hour means**: Mode S is an hour value of many aircraft; the mast is compared as its hour mean, MEPS and the sondes as single steps.
- **The night gap** of the aircraft (about 00–03 UTC).

7.2 Next steps

1. **The cold difference of the aircraft below 200 m**: split by landings and departures, by runway and by aircraft type, from the per-flight low-level profiles.
2. **The mast's 217-m and 327-m directions**: ask for or find FMI's metadata of the booms; if confirmed, correct them in the comparison.
3. **Inversion criterion**: test a Mode S surface inversion from the 400–600-ft bands against the METAR, scored against the mast.
4. **MEPS in stable conditions**: the model's 26-m temperature on calm evenings, over the winter.
5. **Per-flight dots**: each landing's and departure's band wind at the mast's heights beside the hour values (planned in the time series).
6. **Monthly update** of the tables, and a winter case with a deep inversion.

References

[1] Finnish Meteorological Institute. *Open data*: WFS stored query `fmi::observations::weather::mast::multipointcoverage`, station Espoo Latokaski (fmisid 101000). Licence CC BY 4.0. https://en.ilmatieteenlaitos.fi/open-data

[2] de Haan, S. (2011). High-resolution wind and temperature observations from aircraft tracked by Mode-S air traffic control radar. *J. Geophys. Res.* 116, D10111. doi:10.1029/2010JD015264

[3] Müller, M., et al. (2017). AROME-MetCoOp: A Nordic convective-scale operational weather prediction model. *Weather and Forecasting* 32(2), 609–627.

[4] Baas, P., Bosveld, F.C., Klein Baltink, H., Holtslag, A.A.M. (2009). A climatology of nocturnal low-level jets at Cabauw. *J. Appl. Meteor. Climatol.* 48(8), 1627–1642. doi:10.1175/2009JAMC1965.1

[5] Laakso, O. (2026). *Low-Level Windshear Detection from Passively Received Mode S EHS at Helsinki-Vantaa — study log*, version 1.10, 1 November 2026. doc/EFHK_MODE_S_EHS_Windshear_Detection_Study.pdf (the boundary layer: Section "The boundary layer").

[6] Laakso, O. (2026). *The Sounding and Wind Map pages — User guide*. doc/Sounding_Page_User_Guide.pdf.